Project Title:

TravelGuideAI – AI-Based Personalized Travel Itinerary Generator

Submitted by:
Meghana neelam
meghananeelam4954@gmail.com
22P31A4449

1. Abstract :

TravelGuideAI is a web-based AI application that automatically generates personalized travel itineraries based on user inputs such as destination, duration, and interests. The system uses the Gemini 2.5 Flash large language model provided by **Google** to generate structured, day-wise travel plans including tourist attractions, local food suggestions, and travel tips.

The application is built using Python and Streamlit, and is deployed on **Streamlit Community Cloud**. The project demonstrates the integration of generative AI into a real-world application with secure API handling and cloud deployment.

2.Introduction

Planning a trip manually requires researching destinations, organizing daily schedules, finding local attractions, and identifying food and cultural experiences. This process is time-consuming and often confusing.

TravelGuideAI solves this problem by automating itinerary creation using Artificial Intelligence. The user simply enters travel details, and the system generates a structured travel plan within seconds.

The application provides:

- Day-wise planning
- Famous tourist attractions
- Local cuisine recommendations
- Travel tips
- Personalized suggestions based on interests

3. Problem Statement

Traditional travel planning involves:

- Searching multiple websites
- Comparing reviews
- Manually organizing schedules
- Rewriting notes into a structured plan

There is a need for a system that:

- Automates itinerary generation
- Reduces manual effort
- Provides structured and readable travel plans
- Works instantly with minimal input

---

4. Objectives

The main objectives of the project are:

- To build an AI-powered travel planner
- To integrate Gemini API into a web application
- To generate personalized itineraries dynamically
- To deploy the application online for public access
- To ensure secure API key handling

---

5. System Architecture

Workflow Explanation

1. User enters:
    - Destination
    - Number of days
    - Number of nights
    - Interests (optional)
2. Streamlit UI collects input data.
3. A prompt is dynamically generated.
4. The prompt is sent to Gemini 2.5 Flash model via API.

5. The AI processes the request.
6. A structured travel itinerary is generated.
7. The output is displayed in the web interface.

---

6. Technologies Used

Programming Language:

- Python

Frontend Framework:

- Streamlit

AI Model:

- Gemini 2.5 Flash (Google Generative AI)

API Integration:

- google-genai SDK

Deployment Platform:

- **Streamlit Community Cloud**

Version Control:

- **GitHub**

---

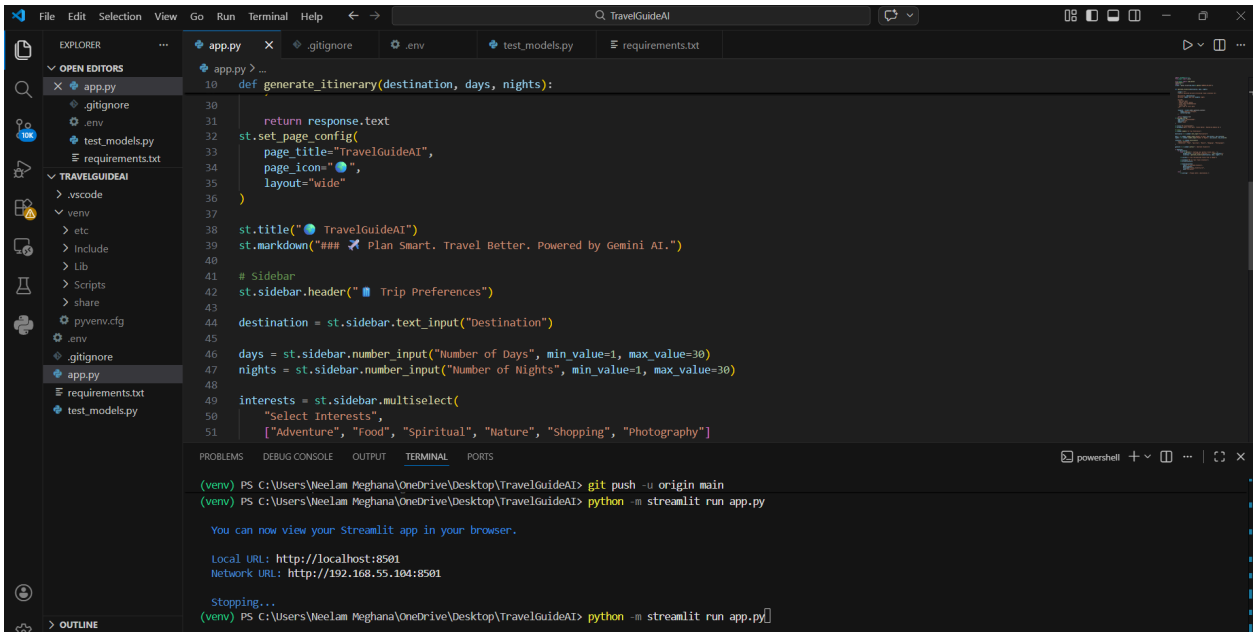
7. Key Features

- User-friendly interface
- AI-generated structured itinerary
- Interest-based customization
- Real-time response generation
- Downloadable itinerary option

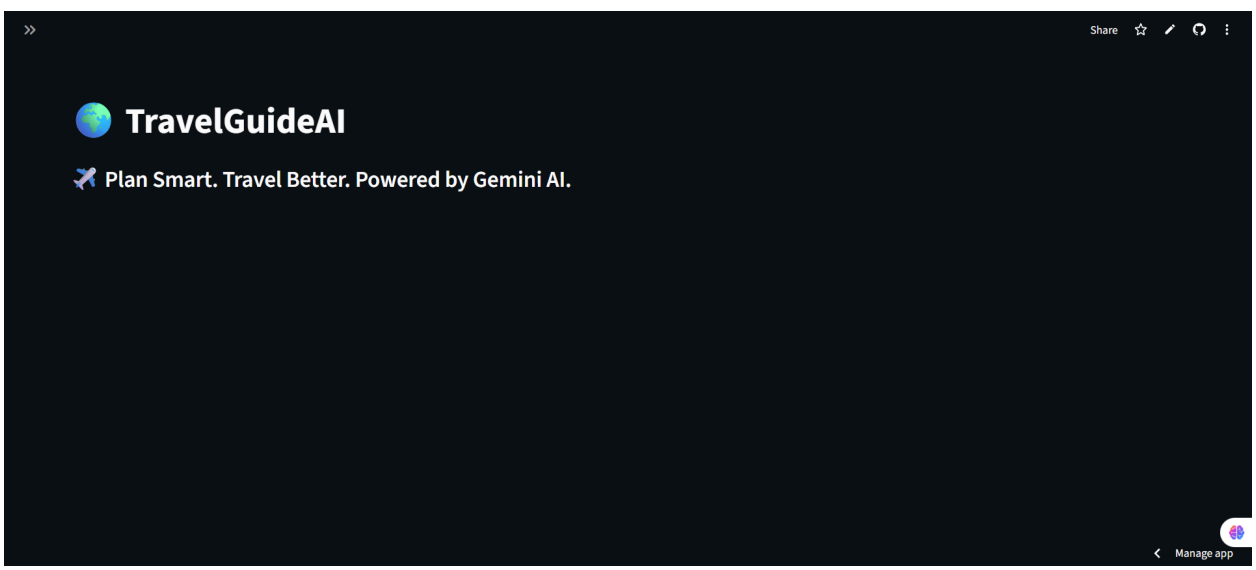
- Secure API key management
- Cloud deployment support

---

8. User Interface Screens:

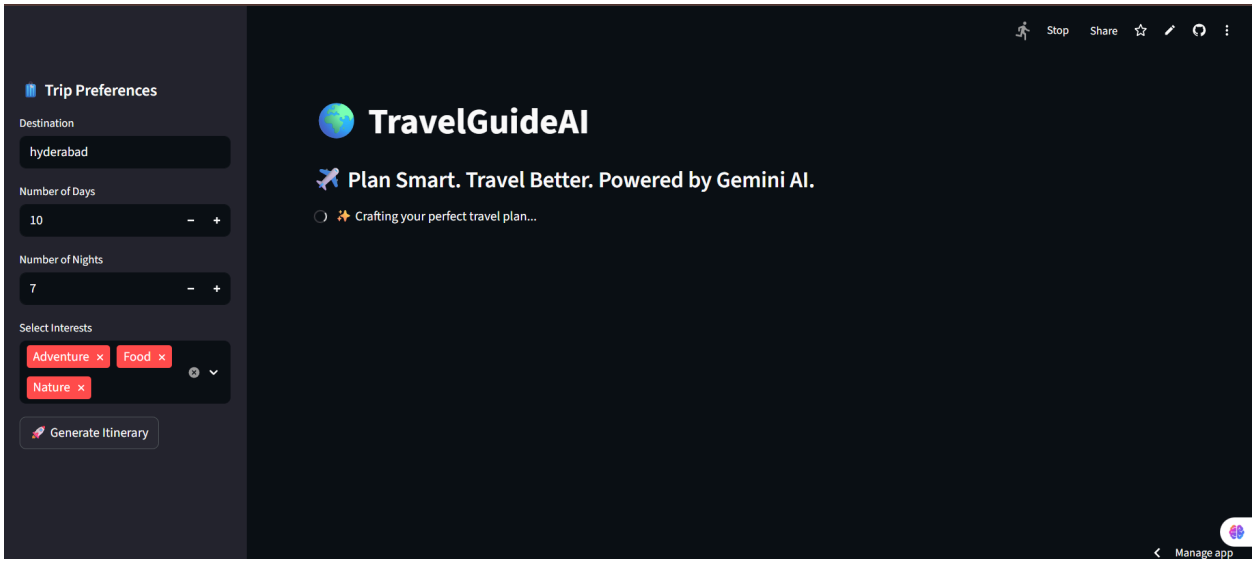


🔹 Home Screen



Description:
The home screen displays the project title and a brief description of the application. The layout is clean and minimal.

🔹 User Input Section



Description:
Users can enter:

- Destination
- Number of days
- Number of nights
- Interests (multi-select option)

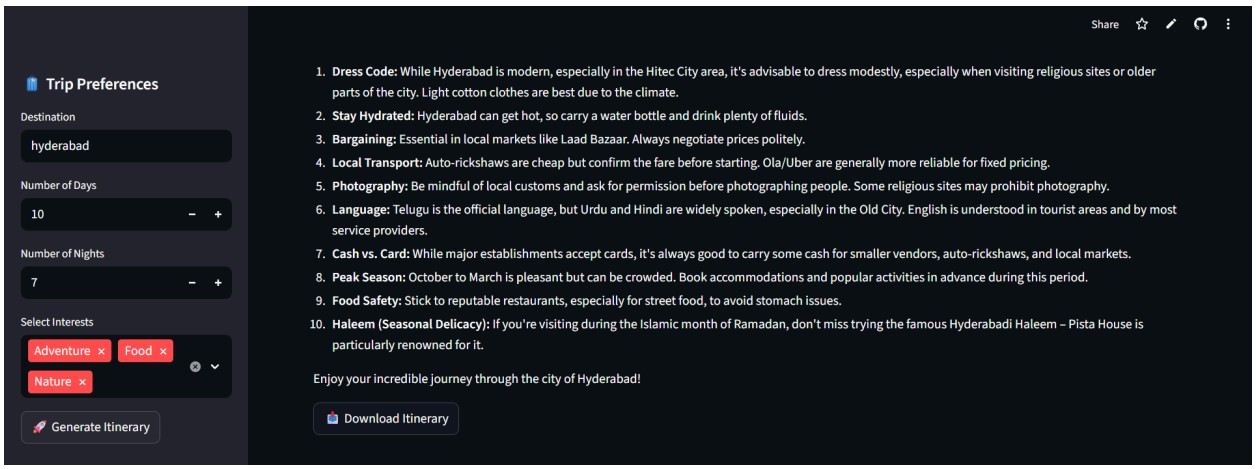
This section is placed in the sidebar for better UI structure.

🔹 Generated Itinerary Output

Description:
After clicking “Generate Itinerary”, the AI-generated travel plan is displayed in a structured format. It includes:

- Day-wise breakdown
- Attractions
- Food recommendations
- Travel tips

🔹 Download Feature



Description:
Users can download the generated itinerary as a text file for future reference.

---

9. Implementation Details

The application initializes the Gemini client using a secure environment variable:

client = genai.Client(api_key=os.getenv("GEMINI_API_KEY"))

The model used:

gemini-2.5-flash

A dynamic prompt is created based on user input and sent to the model using:

client.models.generate_content()

The response is then displayed using Streamlit components.

---

10. Security Implementation

To protect sensitive credentials:

- API key is not hardcoded.
- API key is stored using Streamlit Secrets Manager.
- `.env` file is excluded using `.gitignore`.
- Repository does not contain sensitive data.

---

11. Advantages of the System

- Saves time in travel planning
- Reduces research effort
- Provides organized day-wise schedules
- Easy to use interface
- Free cloud deployment
- Scalable for future enhancements

---

12. Limitations

- Dependent on internet connection
- Limited by API usage quota
- AI output may vary in style
- Does not currently integrate live maps or booking services

---

13. Future Enhancements

- Budget-based travel planning
- Hotel and flight suggestions
- Google Maps integration
- Chat-based conversational travel assistant
- Multi-language support
- Mobile-friendly UI
- PDF export option

14. Conclusion

TravelGuideAI successfully demonstrates the practical application of generative AI in solving real-world problems. By integrating Gemini 2.5 Flash with Streamlit, the system delivers instant, personalized travel itineraries in a clean and accessible interface.

The project highlights skills in:

- AI integration
- API handling
- Web development
- Cloud deployment
- Secure coding practices

This system can be expanded into a commercial travel planning assistant in the future.